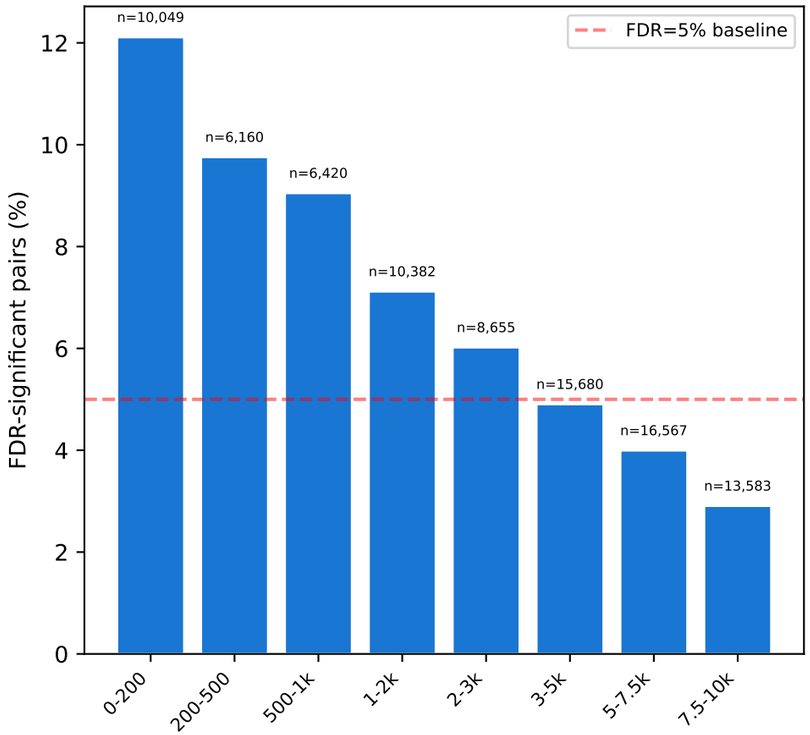
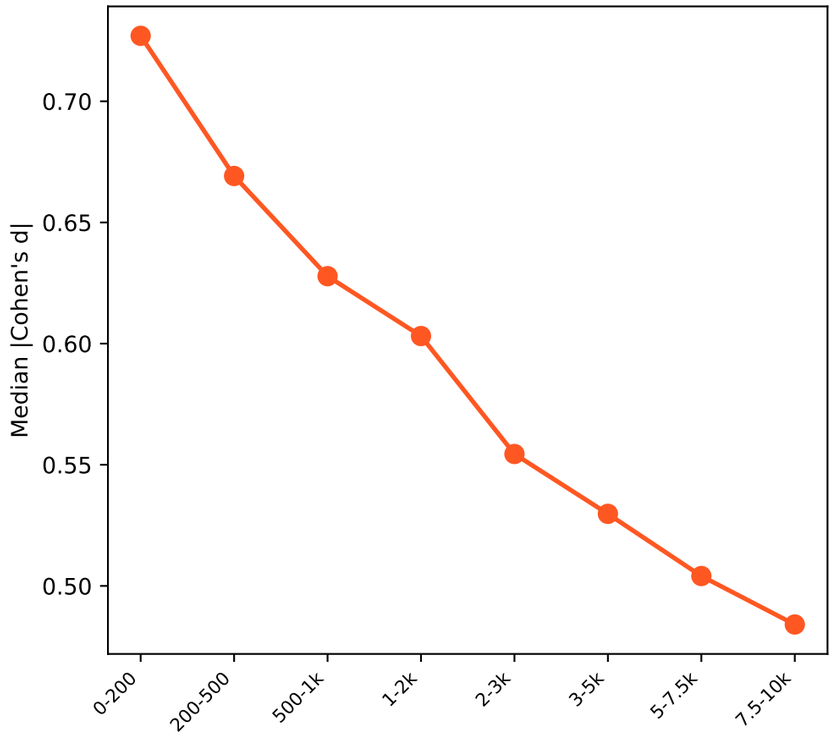
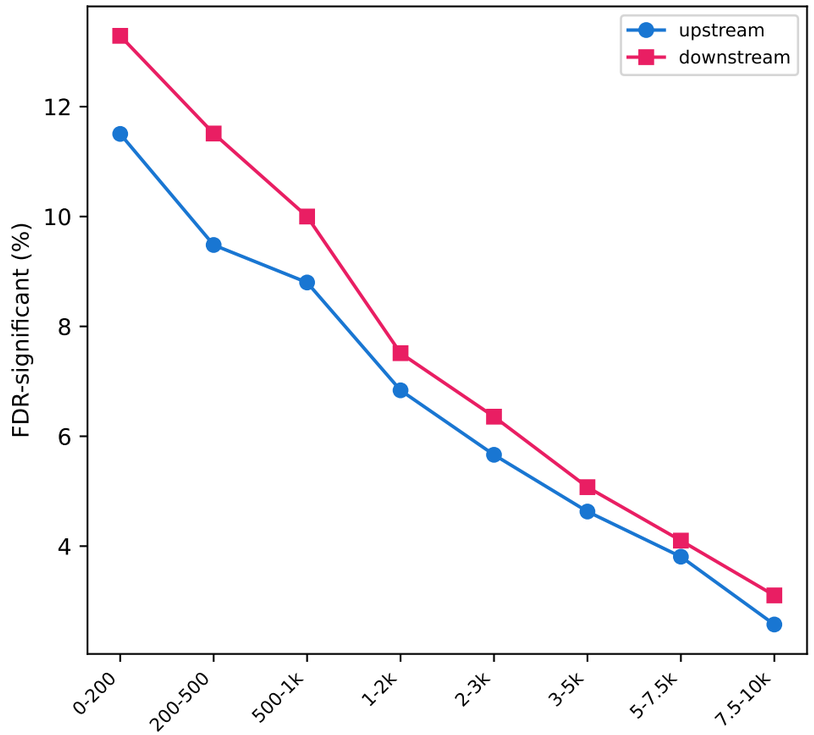
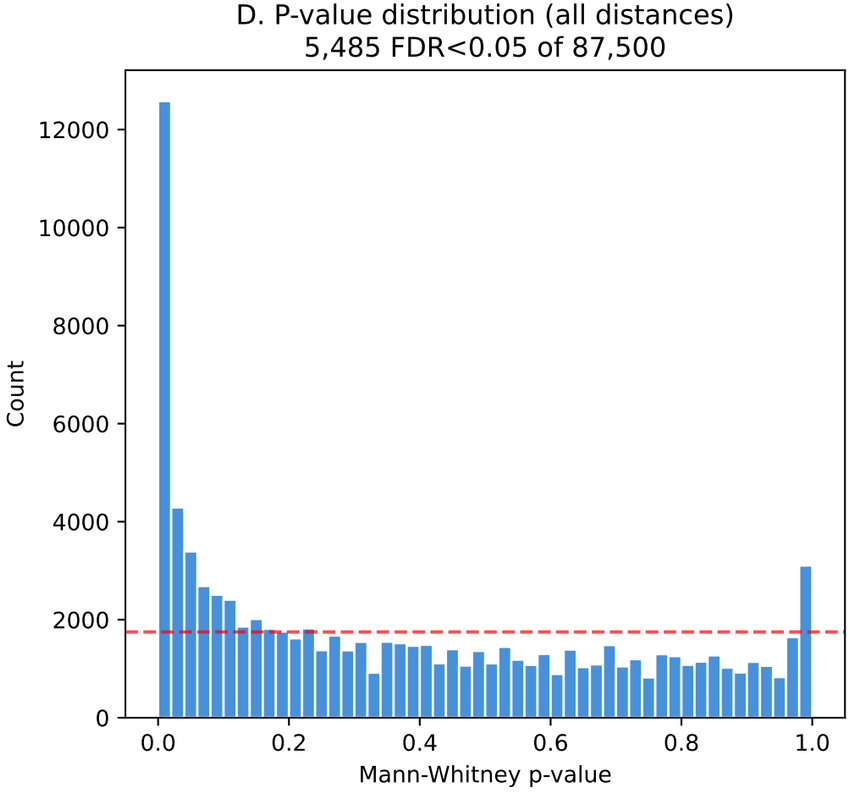
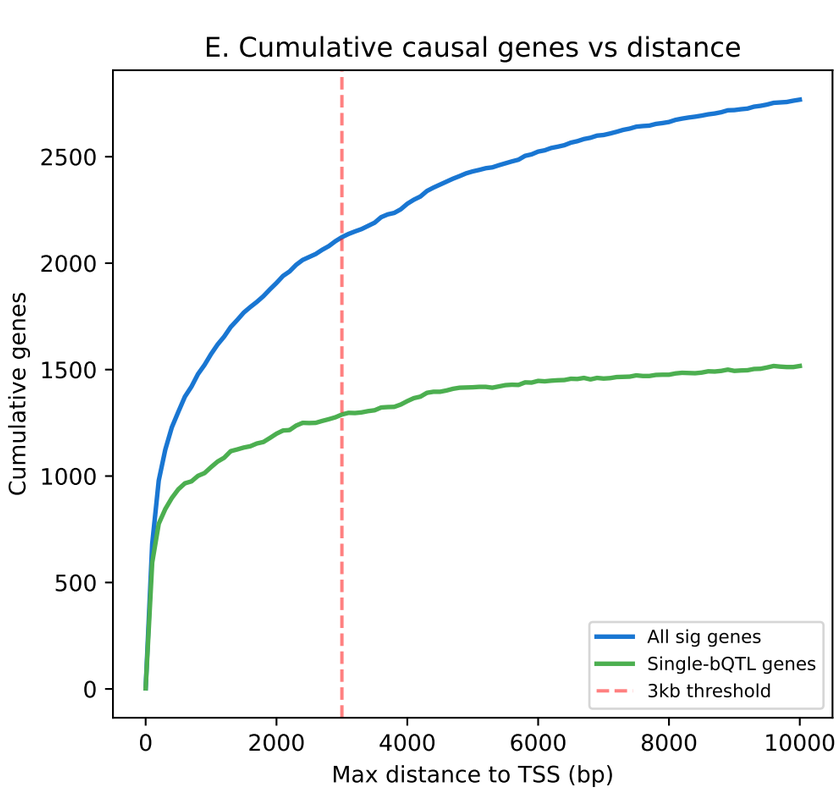
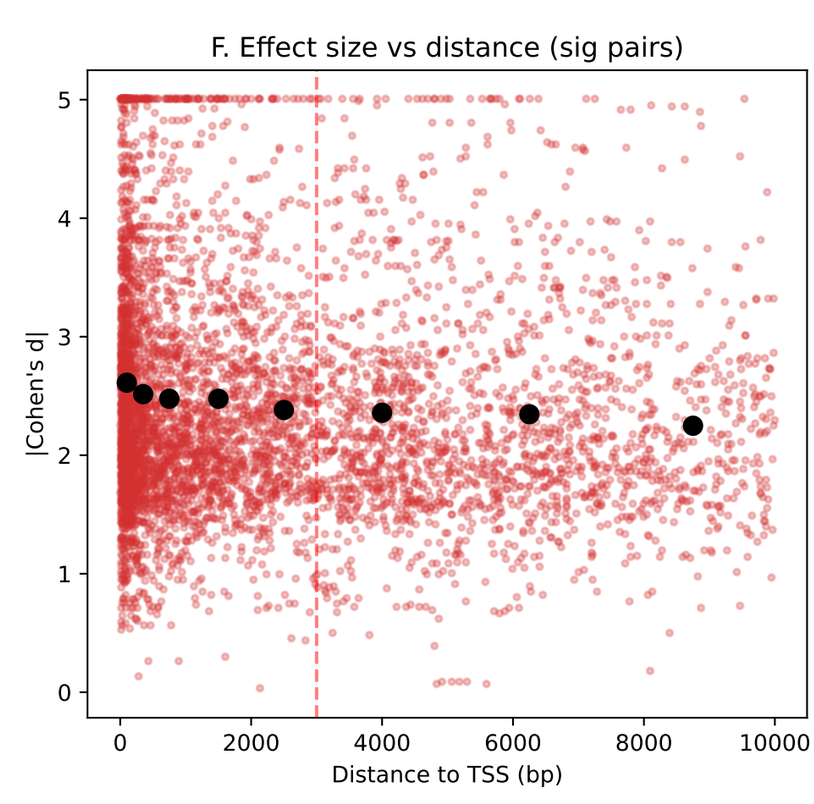

A. Causality rate decays with distance
FDR=5% baseline
n=10,049
n=6,160
n=6,420
n=10,382
n=8,655
n=15,680
n=16,567
n=13,583
FDR-significant pairs (%)
0-200
200-500
500-1k
1-2k
2-3k
3-5k
5-7.5k
7.5-10k


B. Effect size decays with distance
Median |Cohen's d|
0-200
200-500
500-1k
1-2k
2-3k
3-5k
5-7.5k
7.5-10k


C. Upstream vs downstream
upstream
downstream
FDR-significant (%)
0-200
200-500
500-1k
1-2k
2-3k
3-5k
5-7.5k
7.5-10k


D. P-value distribution (all distances)
5,485 FDR<0.05 of 87,500
Count
Mann-Whitney p-value


E. Cumulative causal genes vs distance
Cumulative genes
All sig genes
Single-bQTL genes
3kb threshold
Max distance to TSS (bp)


F. Effect size vs distance (sig pairs)
|Cohen's d|
Distance to TSS (bp)